

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Conducted energy assessments and site surveys across industrial facilities, collecting and analysing energy consumption data using **Python** (pandas, NumPy) and **SQL** queries to identify inefficiencies and quantify energy saving opportunities with **95% data accuracy**.
- Developed automated **data pipelines and ETL workflows** using **Python** to process industrial utility system datasets (HVAC, Steam, Chilled Water, Compressed Air, Cooling Towers, electrical distribution logs), reducing manual data entry by **40%** and enabling real-time energy baseline analysis.
- Prepared technical reports, feasibility studies, engineering calculations, and design documentation for industrial decarbonisation projects; supported Measurement Verification (MV) activities to validate and track post-implementation energy savings against **ISO 50001**, **EN 16247**, and **IS 399** standards.
- Built and maintained operational dashboards using **Power BI** and **Tableau** to visualize energy consumption trends, carbon reduction metrics, and project ROI for industrial clients; created **DAX measures** and **Power Query ETL** models to automate Measurement Verification reporting and stakeholder communication.
- Collaborated across cross-functional teams (senior engineers, sustainability BAs, client operations) to gather requirements, attend site visits, prepare project scopes, bills of quantities, and cost estimates; delivered technical presentations and site reports demonstrating **25% improvement** in stakeholder engagement and UAT sign-off timelines.
- Supported continuous improvement of industrial decarbonisation tools and technical processes through research, benchmarking, and process automation; contributed to internal training materials and documentation aligned with professional chartered development pathways.

SKILLS

Energy & Industrial Systems – Energy assessments, site surveys, HVAC, Steam systems, Chilled Water, LPHW, Compressed Air, Cooling Towers, electrical distribution, industrial utility systems analysis.

Energy Standards & Frameworks – ISO 50001, EN 16247, IS 399, Measurement Verification (MV), energy baseline development, carbon accounting, industrial decarbonisation.

Data Analysis & Engineering – Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, MySQL, Oracle), ETL pipeline development, data collection, energy consumption data analysis, statistical analysis.

Business Intelligence & Reporting – Power BI, Tableau, DAX measures, Power Query, data modelling, energy dashboards, operational reporting, stakeholder reporting.

Technical Documentation & Communication – Feasibility studies, engineering calculations, technical reports, design documentation, bills of quantities, cost estimation, client engagement.

Engineering & Professional Development – Industrial systems design support, process automation, continuous improvement, research and benchmarking, professional chartered alignment.

Additional Technical Skills – Linux, Git, REST APIs, cloud data storage, Docker, Jira, Confluence, strong analytical problem-solving.

EDUCATION AND TRAINING

PROJECTS

Industrial Energy Baseline & Measurement Verification Dashboard (HCL Technologies)

- Engineered end-to-end **Python ETL pipeline** to ingest multi-source industrial utility data (HVAC logs, steam metering, electrical distribution feeds, Compressed Air system readings) from client production environments into a normalized **PostgreSQL** data warehouse.
- Developed interactive **Power BI** and **Tableau** dashboards to calculate energy baselines, track Measurement Verification (MV) metrics, and visualize carbon reduction KPIs in alignment with **ISO 50001** and **EN 16247** standards; automated **DAX**-based variance analysis reduced MV reporting cycle time by **50%**.
- Delivered technical feasibility studies and engineering calculations supporting industrial decarbonisation project scopes; collaborated with client operations teams and senior engineers to validate energy saving opportunities and prepare cost estimates, achieving **98% forecast accuracy** in post-implementation savings validation.

Ireland Jobs Tracker – AI-Driven Energy & Sustainability Project Intelligence (Personal Project)

- Architected **Claude API** (Anthropic) integration using **prompt engineering** and structured JSON workflows to automatically generate tailored project summaries and technical intelligence from energy sector job descriptions and industrial decarbonisation case studies at scale.
- Orchestrated end-to-end automation pipeline: **Apify web scrapers** → data ingestion → **Claude-haiku-4-5** content generation → **reportlab PDF** export → **GitHub Actions** CI → **Vercel** deployment, enabling rapid synthesis of industrial energy data and carbon intelligence for cross-sector analysis.
- Applied hands-on **AI workflow orchestration**, **API integration**, and **prompt engineering** to deliver structured energy and sustainability insights; demonstrated proficiency in modern automation tooling directly applicable to decarbonisation data analysis and industrial reporting workflows.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)